

An Analysis of Text summarization

Text summarization

Text summarization -- Part 1

Diversity Multi-document extractive summarization faces a problem of redundancy. Ideally, we want to extract sentences that are both "central" (i.e., contain the main ideas) and "diverse" (i.e., they differ from one another). For example, in a set of news articles about some event, each article is likely to have many similar sentences. To address this issue, LexRank applies a heuristic post-processing step that adds sentences in rank order, but discards sentences that are too similar to ones already in the summary. This method is called Cross-Sentence Information Subsumption (CSIS). These methods work based on the idea that sentences "recommend" other similar sentences to the reader. Thus, if one sentence is very similar to many others, it will likely be a sentence of great importance. Its importance also stems from the importance of the sentences "recommending" it. Thus, to get ranked highly and placed in a summary, a sentence must be similar to many sentences that are in turn also similar to many other sentences. This makes intuitive sense and allows the algorithms to be applied to an arbitrary new text. The methods are domain-independent and easily portable. One could imagine the features indicating important sentences in the news domain might vary considerably from the biomedical domain. However, the unsupervised "recommendation"-based approach applies to any domain. A related method is Maximal Marginal Relevance (MMR), which uses a general-purpose graph-based ranking algorithm like Page/Lex/TextRank that handles both "centrality" and "diversity" in a unified mathematical framework based on absorbing Markov chain random walks (a random walk where certain states end the walk). The algorithm is called GRASSHOPPER. In addition to explicitly promoting diversity during the ranking process, GRASSHOPPER incorporates a prior ranking (based on sentence position in the case of summarization). The state of the art results for multi-document summarization are obtained using mixtures of submodular functions. These methods have achieved the state of the art results for Document Summarization Corpora, DUC 04 - 07. Similar results were achieved with the use of determinantal point processes (which are a special case of submodular functions) for DUC-04. A new method for multi-lingual multi-document summarization that avoids redundancy generates ideograms to represent the meaning of each sentence in each document, then evaluates similarity by comparing ideogram shape and position. It does not use word frequency, training or preprocessing. It uses two user-supplied parameters: equivalence (when are two sentences to be considered equivalent?) and relevance (how long is the desired summary?).

Text summarization -- Part 2

History The first publication in the area dates back to 1957 (Hans Peter Luhn), starting with a statistical technique. Research increased significantly in 2015. Term frequency-inverse document frequency had been used by 2016. Pattern-based summarization was the most powerful option for multi-document summarization found by 2016. In the following year it was surpassed by latent semantic analysis (LSA) combined with non-negative matrix factorization (NMF). Although they did not replace other approaches and are often combined with them, by 2019 machine learning methods dominated the extractive summarization of single documents, which was considered to be nearing maturity. By 2020, the field was still very active and research is shifting towards abstractive summation and real-time summarization.

Text summarization -- Part 3

Submodular functions as generic tools for summarization The idea of a submodular set function has recently emerged as a powerful modeling tool for various summarization problems. Submodular functions naturally model notions of coverage, information, representation and diversity. Moreover, several important combinatorial optimization problems occur as special instances of submodular optimization. For example, the set cover problem is a special case of submodular optimization, since the set cover function is submodular. The set cover function attempts to find a subset of objects which cover a given set of concepts. For example, in document summarization, one would like the summary to cover all important and relevant concepts in the document. This is an instance of set cover. Similarly, the facility location problem is a special case of submodular functions. The Facility Location function also naturally models coverage and diversity. Another example of a submodular optimization problem is using a determinantal point process to model diversity. Similarly, the Maximum-Marginal-Relevance procedure can also be seen as an instance of submodular optimization. All these important models encouraging coverage, diversity and information are all submodular. Moreover, submodular functions can be efficiently combined, and the resulting function is still submodular. Hence, one could combine one submodular function which models diversity, another one which models coverage and use human supervision to learn a right model of a submodular function for the problem. While submodular functions are fitting problems for summarization, they also admit very efficient algorithms for optimization. For example, a simple greedy algorithm admits a constant factor guarantee. Moreover, the greedy algorithm is extremely simple to implement and can scale to large datasets, which is very important for summarization problems. Submodular functions have achieved state-of-the-art for almost all summarization

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problems. For example, work by Lin and Bilmes, 2012 shows that submodular functions achieve the best results to date on DUC-04, DUC-05, DUC-06 and DUC-07 systems for document summarization. Similarly, work by Lin and Bilmes, 2011, shows that many existing systems for automatic summarization are instances of submodular functions. This was a breakthrough result establishing submodular functions as the right models for summarization problems. Submodular Functions have also been used for other summarization tasks. Tschitschek et al., 2014 show that mixtures of submodular functions achieve state-of-the-art results for image collection summarization. Similarly, Bairi et al., 2015 show the utility of submodular functions for summarizing multi-document topic hierarchies. Submodular Functions have also successfully been used for summarizing machine learning datasets.

Text summarization -- Part 4

"The Army Corps of Engineers, rushing to meet President Bush's promise to protect New Orleans by the start of the 2006 hurricane season, installed defective flood-control pumps last year despite warnings from its own expert that the equipment would fail during a storm, according to documents obtained by The Associated Press". A keyphrase extractor might select "Army Corps of Engineers", "President Bush", "New Orleans", and "defective flood-control pumps" as keyphrases. These are pulled directly from the text. In contrast, an abstractive keyphrase system would somehow internalize the content and generate keyphrases that do not appear in the text, but more closely resemble what a human might produce, such as "political negligence" or "inadequate protection from floods". Abstraction requires a deep understanding of the text, which makes it difficult for a computer system. Keyphrases have many applications. They can enable document browsing by providing a short summary, improve information retrieval (if documents have keyphrases assigned, a user could search by keyphrase to produce more reliable hits than a full-text search), and be employed in generating index entries for a large text corpus. Depending on the different literature and the definition of key terms, words or phrases, keyword extraction is a highly related theme.